



SHARJEEL'S COMPLETE MERN MASTERY ROADMAP

Week 4 → Week 15 | January 4, 2026 → March 20, 2026

TARGET: Devsinc / Systems Limited Junior MERN Developer

CURRENT POSITION: Week 4, Day 22 (January 4, 2026)

DEVICE: Laptop Available

DAILY CAPACITY: 5-6 hours

EXAM BREAKS: Managed, curriculum pauses during university exams



PROGRESS TRACKER (COPY THIS TO NEW CHATS)



COMPLETED:

- Week 1: ES6 Foundations (Dec 2-8, 2025)
- Week 2: Array Methods Mastery (Dec 9-14, 2025)
- Week 3: Async JavaScript (Dec 20-26, 2025) - Gatekeeper PASSED (8.5/10)
- Weather Dashboard: DEPLOYED ✓



CURRENT:

- Week 4, Day 22: Advanced DOM Traversal (January 4, 2026)



UPCOMING:

- Phase 2: React (5 weeks)
 - Phase 3: Backend (3 weeks)
 - Phase 4: Portfolio & Job Prep (3 weeks)
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PHASE-BY-PHASE BREAKDOWN

PHASE 1: JAVASCRIPT MASTERY (WEEKS 1-4) - 90% COMPLETE



WEEK 1: ES6 FOUNDATIONS (COMPLETED)

Dates: December 2-8, 2025

Status: ✓ Complete + Exit Exam Passed

Topics Covered:

- Day 1: Scope (var/let/const)
- Day 2: Arrow Functions
- Day 3: Template Literals
- Day 4: Destructuring & Spread
- Day 5: ES6 Modules
- Day 6-7: Console ToDo List Project

Key Takeaway: Solid ES6+ syntax foundation

 WEEK 2: ARRAY METHODS MASTERY (COMPLETED)

Dates: December 9-14, 2025

Status:  Complete + Exit Exam Passed

Topics Covered:

- Day 8: forEach & for...of
- Day 9: .map() - Transformation
- Day 10: .filter() - Selection
- Day 11: .reduce() - Aggregation
- Day 12: Object.keys/values/entries
- Day 13-14: Shopping Cart Logic System

Key Takeaway: Functional programming patterns mastered

 WEEK 3: ASYNCHRONOUS JAVASCRIPT (COMPLETED)

Dates: December 20-26, 2025

Status:  Complete + Gatekeeper Exam PASSED (8.5/10) 🎉

Topics Covered:

- Day 15: Callbacks & Callback Hell
- Day 16: Promises (.then/.catch/.finally)
- Day 17: Async/Await + Try/Catch
- Day 18: Fetch API + JSONPlaceholder
- Day 19: localStorage Basics
- Day 20-21: Weather Dashboard (DEPLOYED ✓)

Gatekeeper Questions:

1. Callback vs Promise vs Async/Await comparison
2. Live coding: Restaurant filtering with async
3. Real-world architecture for API workflows

Key Takeaway: Event loop mastery, production-ready async patterns

WEEK 4: ADVANCED JAVASCRIPT & DOM MASTERY (IN PROGRESS)

Dates: January 4-10, 2026

Status:  Day 22 Starting Today

Goal: Complete JavaScript foundations + Phase 1 Capstone

Day 22 - January 4 (Saturday) ← YOU ARE HERE

Topic: Advanced DOM Traversal & Manipulation

Time: 5-6 hours

Device: Laptop

Learning Objectives:

- Master parent/child/sibling navigation
- Use closest() for ancestor lookup
- Implement nested comment system logic
- Understand DOM tree relationships

Project: Nested Comment System Simulator

Deliverable: `day22-dom-traversal.js`

Day 23 - January 5 (Sunday)

Topic: Event Bubbling & Event Delegation

Time: 5-6 hours

Device: Laptop

Learning Objectives:

- Understand event propagation
- Implement single-listener pattern for multiple elements
- Apply delegation to dynamic content
- Optimize performance with event delegation

Project: Lahore Restaurants Social Feed

Deliverables:

- `index.html`
- `style.css`
- `day23-delegation.js`

Key Feature: 1 listener handling 50+ post interactions

Day 24 - January 6 (Monday)

Topic: Form Validation (Pure JavaScript)

Time: 5-6 hours

Device: Laptop

Learning Objectives:

- Validate email with regex
- Implement password strength meter
- Real-time feedback on user input
- Enable/disable submit based on validation state

Project: Careem-Style Signup Form

Deliverables:

- `day24-validation.html`
- `day24-validation.css`
- `day24-validation.js`

Validation Rules:

- Email format check
 - Pakistan phone number (03XX-XXXXXXX)
 - Password strength (weak/medium/strong)
 - Confirm password match
 - Terms checkbox required
-

Day 25 - January 7 (Tuesday)

Topic: Chrome DevTools & Systematic Debugging

Time: 5-6 hours

Device: Laptop

Learning Objectives:

- Use breakpoints effectively
- Inspect variables during execution
- Monitor network requests
- Apply systematic debugging workflow

Project: Debug Challenge - Fix 10 Broken Functions

Deliverable: `day25-debugging-challenge.js`

Bug Types:

- Off-by-one errors
 - Wrong comparison operators
 - Missing return statements
 - Async without await
 - Scope errors (var vs let)
 - Array method misuse
-

Day 26 - January 8 (Wednesday)

Topic: Object-Oriented Programming with ES6 Classes

Time: 5-6 hours

Device: Laptop

Learning Objectives:

- Create classes with constructor
- Implement methods and properties
- Use inheritance (extends)
- Apply OOP to real-world scenarios

Project: Lahore Transport System (OOP)

Deliverable: `day26-oop-transport.js`

Classes:

- `Vehicle` (base class)
- `Rickshaw` extends `Vehicle`
- `Careem` extends `Vehicle`
- `MetroBus` extends `Vehicle`
- `TransportSystem` (management)

Features:

- Book rides
 - Calculate fares
 - Track earnings
 - Generate daily reports
-

Day 27-28 - January 9-10 (Thursday-Friday)

Topic: PHASE 1 CAPSTONE PROJECT

Time: 12-16 hours total (6-8 per day)

Device: Laptop

Project: 🎬 **Movie Search Application**

Tech Stack:

- HTML5/CSS3/JavaScript (ES6+)
- OMDB API (movie database)
- localStorage for favorites
- Responsive design

Core Features:

1. Movie search with OMDB API
2. Display results with posters

3. Add/remove favorites (localStorage)
4. Tab switching (Results/Favorites)
5. Search validation
6. Error handling
7. Loading states

Advanced Features:

1. Movie details modal (click to expand)
2. Search history (last 5 searches)
3. Filter by type (movie/series)
4. Sort by year
5. Export favorites as JSON

Day 27 Focus:

- API integration
- Core CRUD functionality
- Favorites system with localStorage
- Basic UI/UX

Day 28 Focus:

- Advanced features (modal, filters, sorting)
- Polish & responsive design
- Professional README
- Deployment preparation

Deliverables:

- `index.html`
- `style.css`
- `app.js`
- `README.md` (comprehensive project documentation)

Success Criteria:

- All features working
- Professional appearance
- Deployed on Vercel/Netlify
- Posted on LinkedIn
- Ready for portfolio

WEEK 4 EXIT CRITERIA (NO FORMAL EXAM)

To advance to Phase 2 (React):

Technical Competence:

- Movie app fully functional with 8+ features

- All Day 22-26 projects committed to GitHub
- Can explain event delegation benefits
- Understands OOP class structure
- Comfortable with Chrome DevTools

✅ **Portfolio Quality:**

- Movie app deployed with live link
- Professional README files
- Clean, commented code
- No major bugs

✅ **Self-Assessment:**

- Can teach DOM traversal to junior
- Can debug systematically without panic
- Understands when to use classes vs plain objects

IF PASSED: Week 5 React begins January 13! 🎉

IF GAPS: Address specific areas before proceeding

PHASE 2: REACT FRONTEND DEVELOPMENT (WEEKS 5-9)

WEEK 5: REACT FUNDAMENTALS & COMPONENT THINKING

Dates: January 13-19, 2026

Goal: Understand React philosophy and component-based architecture

Day 29 - January 13 (Monday)

Topic: React Introduction & Setup

Learning Objectives:

- Understand React vs Vanilla JS
- Set up Vite project (not Create React App)
- Understand JSX syntax
- Create first component

Project: Hello React - Counter App

Key Concepts:

- Virtual DOM explanation

- Component structure
 - JSX rules and differences from HTML
-

Day 30 - January 14 (Tuesday)

Topic: Components & Props

Learning Objectives:

- Create reusable components
- Pass props to components
- Props vs State distinction
- Component composition patterns

Project: Product Card Grid (Lahore E-commerce)

Features:

- Reusable ProductCard component
 - Props: name, price, image, rating
 - Conditional rendering (in stock/out of stock)
-

Day 31 - January 15 (Wednesday)

Topic: State with useState Hook

Learning Objectives:

- Understand React state
- Use useState hook
- Handle state updates
- Avoid common useState pitfalls

Project: Interactive ToDo List (React Version)

Features:

- Add/delete tasks
 - Mark as complete
 - Filter (All/Active/Completed)
-

Day 32 - January 16 (Thursday)

Topic: Event Handling in React

Learning Objectives:

- Handle onClick, onChange, onSubmit
- Pass parameters to event handlers
- Synthetic events explanation

- Controlled components

Project: Contact Form with Validation

Features:

- Real-time validation
 - Controlled inputs
 - Submit handler
 - Error messages
-

Day 33 - January 17 (Friday)

Topic: Lists & Keys in React

Learning Objectives:

- Render arrays with `.map()`
- Understand key prop importance
- Avoid index as key
- Conditional rendering patterns

Project: Restaurant Menu Display

Features:

- Map through menu items
 - Filter by category
 - Search functionality
 - Dynamic rendering
-

Day 34-35 - January 18-19 (Weekend)

Project: React Shopping Cart

Features:

- Product listing with `.map()`
- Add to cart with `useState`
- Quantity controls
- Total calculation
- Remove from cart

Deliverables:

- Multi-component architecture
 - Props drilling example
 - State management at parent level
 - Professional UI with CSS modules
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WEEK 6: STATE MANAGEMENT & HOOKS MASTERY 🚧 GATEKEEPER

Dates: January 20-26, 2026

Goal: Master React Hooks + Pass MANDATORY Gatekeeper Exam

Day 36 - January 20 (Monday)

Topic: useEffect Hook - Side Effects

Learning Objectives:

- Understand component lifecycle
- Use useEffect for side effects
- Dependency array rules
- Cleanup functions

Project: Live Clock & Weather Widget

Key Concepts:

- API calls in useEffect
 - Interval management
 - Cleanup to prevent memory leaks
-

Day 37 - January 21 (Tuesday)

Topic: useEffect Dependencies Deep Dive

Learning Objectives:

- Understand dependency array thoroughly
- Avoid infinite loops
- Fetch data on mount
- Conditional effects

Project: GitHub User Search

Features:

- Fetch user on search
 - Loading states
 - Error handling
 - Display user repos
-

Day 38 - January 22 (Wednesday)

Topic: Custom Hooks

Learning Objectives:

- Create reusable custom hooks

- Extract logic from components
- Naming convention (use prefix)
- Share stateful logic

Project: useFetch, useLocalStorage, useToggle hooks

Application: Recipe Search App using custom hooks

Day 39 - January 23 (Thursday)

Topic: useReducer for Complex State

Learning Objectives:

- Understand reducer pattern
- When to use useReducer vs useState
- Actions and action creators
- Complex state logic

Project: Task Manager with useReducer

Features:

- Add/edit/delete tasks
 - Filter and sort
 - Undo/redo functionality
-

Day 40 - January 24 (Friday)

Topic: useRef & DOM Access

Learning Objectives:

- Access DOM elements directly
- Persist values across renders
- Focus management
- Difference from useState

Project: Auto-focus Input & Video Player Controls

Day 41-42 - January 25-26 (Weekend)

 **WEEK 6 GATEKEEPER EXAMINATION** 

MANDATORY 3-PART EXAM (BLOCKS WEEK 7)

Part 1: Conceptual Deep Dive (30 min)

- Explain useState vs useReducer with real scenarios
- When does useEffect run? (all dependency cases)

- Custom hooks: Why and how?
- Virtual DOM and reconciliation

Part 2: Live Coding Challenge (60 min)

Build: **Blog Post Manager**

- Fetch posts from API (useEffect)
- Search/filter (useState)
- Loading & error states
- Custom useFetch hook
- Display with proper keys

Requirements:

- Must use useState, useEffect, custom hook
- Error handling required
- Professional UI
- Deploy to Vercel

Part 3: Debug Challenge (30 min)

Fix 5 intentionally broken React components:

- Infinite useEffect loop
- Missing dependency
- Wrong key usage
- State update timing issue
- useEffect cleanup missing

PASSING CRITERIA:

- Part 1: 8/10 minimum (conceptual clarity)
- Part 2: Fully functional deployed app
- Part 3: Fix at least 4/5 bugs correctly

IF PASSED:

"Mashallah Sharjeel! React Hooks conquered. Week 7 Advanced Patterns unlocked. You're now thinking in React - this is where 70% of bootcamp students fail. You're in the top tier."

IF FAILED:

"Week 6 repeat karna hoga. Hooks React ki rooh hain. Aage Context API, Redux - sab hooks pe depend karte hain. Foundation perfect hona chahiye. Main tumhare saath hoon."

WEEK 7: ADVANCED REACT PATTERNS & STYLING

Dates: January 27 - February 2, 2026

Prerequisite: Week 6 Gatekeeper PASSED 

Day 43 - January 27 (Monday)

Topic: Component Composition Patterns

Learning Objectives:

- Container/Presentational pattern
- Compound components
- Render props pattern
- Children prop advanced usage

Project: Reusable Modal Component System

Day 44 - January 28 (Tuesday)

Topic: React.memo & Performance Optimization

Learning Objectives:

- Understand re-render behavior
- Use React.memo effectively
- useMemo and useCallback hooks
- When NOT to optimize

Project: Optimize Large List Rendering (1000+ items)

Day 45 - January 29 (Wednesday)

Topic: Styling in React (CSS Modules + Tailwind)

Learning Objectives:

- CSS Modules for scoped styles
- Tailwind CSS integration
- Styled-components basics
- When to use which approach

Project: Professional Landing Page with Tailwind

Day 46 - January 30 (Thursday)

Topic: Forms with React Hook Form

Learning Objectives:

- Install and use React Hook Form
- Validation with Yup/Zod
- Error handling
- Performance benefits

Project: Multi-step Registration Form

Day 47 - January 31 (Friday)

Topic: React Dev Tools Mastery

Learning Objectives:

- Component tree inspection
- Props/state debugging
- Performance profiling
- Time-travel debugging

Project: Debug Session - Profile existing apps

Day 48-49 - February 1-2 (Weekend)

Project: E-Commerce Product Catalog

Features:

- Product grid with filters
- Search functionality
- Sorting (price, rating)
- Pagination
- Professional UI with Tailwind
- Optimized with React.memo

Tech:

- React Hook Form for filters
 - Custom hooks for data fetching
 - CSS Modules + Tailwind
 - Deployed on Vercel
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WEEK 8: REACT ROUTER & NAVIGATION

Dates: February 3-9, 2026

Day 50 - February 3 (Monday)

Topic: React Router Basics

Learning Objectives:

- Install React Router v6
- Set up BrowserRouter
- Create routes with Route component

- Navigate with Link component

Project: Multi-page Portfolio Site

Day 51 - February 4 (Tuesday)

Topic: Dynamic Routes & URL Parameters

Learning Objectives:

- useParams hook
- Dynamic segments
- Nested routes
- 404 pages

Project: Blog with Individual Post Pages

Day 52 - February 5 (Wednesday)

Topic: Programmatic Navigation

Learning Objectives:

- useNavigate hook
- Redirect after form submission
- Navigate with state
- Replace vs push

Project: Login Flow with Protected Routes

Day 53 - February 6 (Thursday)

Topic: Nested Routes & Layouts

Learning Objectives:

- Outlet component
- Shared layouts
- Index routes
- Route nesting patterns

Project: Dashboard with Sidebar Navigation

Day 54 - February 7 (Friday)

Topic: Search Params & Query Strings

Learning Objectives:

- useSearchParams hook
- Filter products via URL
- Maintain state in URL
- Back button behavior

Project: Filterable Product Catalog with URL state

Day 55-56 - February 8-9 (Weekend)

Project: Full Blog Application with Router

Features:

- Home, About, Blog, Contact pages
- Individual blog post pages (dynamic routes)
- Category filtering (search params)
- Pagination with URL state
- Protected admin area
- 404 page

Tech:

- React Router v6
 - Context API for auth
 - Tailwind UI
 - Deployed with routing support
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WEEK 9: CONTEXT API & GLOBAL STATE

Dates: February 10-16, 2026

Day 57 - February 10 (Monday)

Topic: Context API Fundamentals

Learning Objectives:

- Understand props drilling problem
- Create Context with createContext
- Provide context value
- Consume with useContext

Project: Theme Switcher (Light/Dark Mode)

Day 58 - February 11 (Tuesday)

Topic: Custom Context Hooks Pattern

Learning Objectives:

- Create custom provider components
- Build useAuth, useTheme hooks
- Organize context code
- Error handling for missing provider

Project: Authentication Context System

Day 59 - February 12 (Wednesday)

Topic: Multiple Contexts & Composition

Learning Objectives:

- Combine multiple contexts
- Avoid context hell
- Context composition patterns
- Performance considerations

Project: App with Auth + Theme + Language contexts

Day 60 - February 13 (Thursday)

Topic: Context with useReducer (Advanced State)

Learning Objectives:

- Combine Context + useReducer
- Dispatch actions globally
- Complex state management
- When to use vs Redux

Project: Shopping Cart with Context + Reducer

Day 61 - February 14 (Friday)

Topic: Context Performance Optimization

Learning Objectives:

- Avoid unnecessary re-renders
- Split contexts strategically
- Memoize context values
- Use separate contexts for data vs actions

Project: Optimize large app with multiple contexts

Day 62-63 - February 15-16 (Weekend)

PHASE 2 CAPSTONE: Full E-Commerce App

Features:

- Product listing with search/filter
- Shopping cart (Context + Reducer)
- User authentication (Context)
- Theme toggle (Context)
- React Router navigation
- Checkout flow
- Order history

Tech Stack:

- React + Vite
- React Router v6
- Context API (auth, cart, theme)
- useReducer for cart logic
- Tailwind CSS
- React Hook Form
- localStorage persistence
- Deployed on Vercel

Success Criteria:

- All features functional
 - Professional UI/UX
 - No prop drilling
 - Optimized performance
 - Mobile responsive
 - Live deployment
 - Comprehensive README
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PHASE 3: BACKEND & DATABASE (WEEKS 10-12)

WEEK 10: NODE.JS & EXPRESS FUNDAMENTALS

Dates: February 17-23, 2026

Goal: Backend development basics with Node.js

Day 64 - February 17 (Monday)

Topic: Node.js Introduction & Setup

Learning Objectives:

- Understand Node.js runtime
- Difference from browser JavaScript
- CommonJS vs ES Modules
- npm package management

Project: Simple HTTP Server (raw Node.js)

Day 65 - February 18 (Tuesday)

Topic: Express.js Basics

Learning Objectives:

- Install and set up Express
- Create routes (GET, POST, PUT, DELETE)
- Request/Response objects
- Middleware concept

Project: REST API - Lahore Restaurants CRUD

Day 66 - February 19 (Wednesday)

Topic: Middleware Deep Dive

Learning Objectives:

- Built-in middleware (express.json, static)
- Custom middleware
- Error handling middleware
- Middleware order importance

Project: Authentication Middleware System

Day 67 - February 20 (Thursday)

Topic: RESTful API Design

Learning Objectives:

- REST principles
- Resource naming conventions
- HTTP status codes (200, 201, 404, 500)
- API versioning

Project: Task Manager API (full CRUD)

Day 68 - February 21 (Friday)

Topic: Environment Variables & Security

Learning Objectives:

- dotenv package
- Secure sensitive data
- CORS setup
- Helmet for security headers

Project: Secure API with environment config

Day 69-70 - February 22-23 (Weekend)

Project: Blog API with Express

Features:

- Posts CRUD (GET, POST, PUT, DELETE)
- Comments CRUD
- Search/filter endpoints
- Pagination
- Error handling
- Validation (express-validator)
- Security middleware
- Documented with Postman

Deliverables:

- Express server
 - 10+ routes
 - Postman collection
 - README with API documentation
-

WEEK 11: MONGODB & MONGOOSE

Dates: February 24 - March 2, 2026

Day 71 - February 24 (Monday)

Topic: MongoDB Introduction

Learning Objectives:

- Understand NoSQL vs SQL
- Install MongoDB locally / Atlas setup
- MongoDB Compass GUI
- Basic CRUD operations in shell

Project: Practice MongoDB queries in Compass

Day 72 - February 25 (Tuesday)

Topic: Mongoose Basics

Learning Objectives:

- Install Mongoose
- Connect to MongoDB
- Create schemas and models
- Basic CRUD with Mongoose

Project: User Model with Validation

Day 73 - February 26 (Wednesday)

Topic: Mongoose Schemas & Validation

Learning Objectives:

- Schema types and validators
- Custom validation
- Virtual properties
- Timestamps

Project: Product Schema with Advanced Validation

Day 74 - February 27 (Thursday)

Topic: Mongoose Relationships

Learning Objectives:

- Referenced documents (population)
- Embedded documents
- One-to-many relationships
- Many-to-many relationships

Project: Blog with Posts-Comments-Users relationships

Day 75 - February 28 (Friday)

Topic: Mongoose Queries & Aggregation

Learning Objectives:

- Find with filters
- Sorting and pagination
- Aggregation pipeline
- Performance optimization

Project: Analytics Dashboard API (aggregation)

Day 76-77 - March 1-2 (Weekend)

Project: Full CRUD API with MongoDB

Features:

- Express + MongoDB + Mongoose
- User authentication (bcrypt)
- JWT tokens
- Protected routes
- File upload (Multer)
- Data validation
- Error handling
- Deployed on Railway

Models:

- User (auth)
 - Product (inventory)
 - Order (transactions)
-

WEEK 12: MERN INTEGRATION 🚧 GATEKEEPER

Dates: March 3-9, 2026

Goal: Connect React frontend with Node/Express/MongoDB backend

Day 78 - March 3 (Monday)

Topic: CORS & API Integration

Learning Objectives:

- Set up CORS in Express

- Fetch data from React to Express
- Handle async state in frontend
- Error handling across stack

Project: Connect React app to Express API

Day 79 - March 4 (Tuesday)

Topic: Authentication Flow (JWT)

Learning Objectives:

- JWT creation and verification
- Login/Register endpoints
- Store token in localStorage
- Protected React routes

Project: Full Auth System (Register/Login/Logout)

Day 80 - March 5 (Wednesday)

Topic: CRUD Operations Full Stack

Learning Objectives:

- Create resource from React form
- Display data from MongoDB
- Update and delete operations
- Optimistic UI updates

Project: Task Manager (MERN)

Day 81 - March 6 (Thursday)

Topic: File Upload & Image Handling

Learning Objectives:

- Multer for backend uploads
- FormData in React
- Image preview before upload
- Serve static files

Project: Profile Picture Upload System

Day 82 - March 7 (Friday)

Topic: Deployment Preparation

Learning Objectives:

- Build React for production
- Serve React from Express
- Environment variables for production
- Database connection strings

Project: Deploy full MERN app (Vercel + Railway)

Day 83-84 - March 8-9 (Weekend)

🚨 WEEK 12 GATEKEEPER: FULL MERN PROJECT 🚨

Build: Task Management SaaS Application

MANDATORY Requirements:

Backend (Express + MongoDB):

- User registration/login (JWT)
- CRUD for tasks (title, description, status, priority)
- Search and filter tasks
- Pagination (10 tasks per page)
- User can only see their tasks
- Input validation
- Error handling

Frontend (React):

- Login/Register pages
- Protected dashboard (redirect if not logged in)
- Task list with add/edit/delete
- Search and filter UI
- Loading states
- Error messages
- Responsive design

Deployment:

- Frontend: Vercel
- Backend: Railway
- Database: MongoDB Atlas

Evaluation Criteria:

Part 1: Code Quality (40%)

- Clean, organized code

- Proper error handling
- Security (no hardcoded secrets)
- Comments where needed

Part 2: Functionality (40%)

- All CRUD operations work
- Authentication flow correct
- Search/filter functional
- Pagination works

Part 3: Professional Polish (20%)

- Professional UI
- Good UX (loading states, errors)
- README documentation
- Live deployment

Time Limit: 12-16 hours (Weekend)

PASSING CRITERIA:

- Score 70%+ overall
- Deployed and functional
- No critical bugs

IF PASSED:

"Incredible Sharjeel! Full MERN stack conquered. You can now build production-ready applications. Week 13 starts your Hero Project - this is what gets you hired."

IF FAILED:

"Backend integration is complex. Week 12 repeat needed. This gatekeeper mirrors real job tasks. Once passed, you're job-ready. Main tumhare saath hoon."

PHASE 4: PORTFOLIO & JOB PREPARATION (WEEKS 13-15)

WEEK 13: HERO PROJECT DEVELOPMENT

Dates: March 10-16, 2026

Goal: Build unique project that solves real problem

Day 85-86 - March 10-11 (Mon-Tue)

Topic: Hero Project Planning

Activities:

- Brainstorm 5 project ideas (Lahore-specific problems)
- Research existing solutions
- Choose unique angle
- Define MVP features
- Create wireframes
- Plan tech stack

Examples:

- Lahore Event Finder (concerts, meetups)
- Skill Exchange Platform (teach/learn locally)
- Local Service Marketplace
- Community Problem Reporter
- Study Group Matcher

Deliverables:

- Project proposal document
 - Feature list (MVP + Nice-to-have)
 - Database schema design
 - API endpoint planning
 - UI wireframes (Figma/hand-drawn)
-

Day 87-89 - March 12-14 (Wed-Fri)

Topic: Hero Project Backend Development

Tasks:

- Set up Express + MongoDB
- Create all models
- Implement authentication
- Build all API endpoints
- Test with Postman
- Deploy backend to Railway

Daily Goals:

- Day 87: Setup + Auth + Models
 - Day 88: CRUD endpoints + Validation
 - Day 89: Advanced features + Deployment
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Day 90-91 - March 15-16 (Weekend)

Topic: Hero Project Frontend Development

Tasks:

- Create React app with Vite
- Set up routing
- Build all pages
- Connect to backend API
- Polish UI with Tailwind
- Add loading/error states
- Deploy to Vercel

Success Criteria:

- Fully functional
 - Professional design
 - Mobile responsive
 - Deployed live
 - Comprehensive README
-

WEEK 14: POLISH & DEPLOYMENT MASTERY

Dates: March 17-23, 2026

Day 92 - March 17 (Monday)

Topic: Hero Project Polish

Tasks:

- Fix all bugs
 - Improve UI/UX
 - Add animations
 - Optimize performance
 - Test on mobile
-

Day 93 - March 18 (Tuesday)

Topic: Professional README & Documentation

Learning:

- Write clear project description
- Document features with screenshots
- Installation instructions
- API documentation
- Technology stack explanation
- Future enhancements

Task: Create portfolio-grade README for Hero Project

Day 94 - March 19 (Wednesday)

Topic: GitHub Profile Optimization

Tasks:

- Professional bio
 - Pinned repositories (top 6 projects)
 - Contribution graph (green squares)
 - Professional profile README
 - Links to deployed projects
-

Day 95 - March 20 (Thursday)

Topic: Deploy All Portfolio Projects

Ensure Deployed:

1. Weather App
2. Movie Search App
3. E-Commerce (React)
4. Task Manager (MERN)
5. Hero Project

Tasks:

- Verify all links work
 - Custom domain (optional)
 - SSL certificates
 - Performance optimization
-

Day 96 - March 21 (Friday)

Topic: Portfolio Website Development

Build: Personal Portfolio Site

Features:

- About section
- Skills showcase
- Projects gallery (deployed links)
- Contact form (EmailJS)
- Blog (optional)
- Responsive design

- Professional appearance

Tech: React + Tailwind + Vercel

Day 97-98 - March 22-23 (Weekend)

Topic: Portfolio Final Polish + Case Studies

Tasks:

- Add project case studies
 - Write blog post about Hero Project
 - Create demo videos (optional)
 - LinkedIn post for each major project
 - Get feedback and improve
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WEEK 15: JOB HUNT MATERIALS & APPLICATIONS

Dates: March 24-30, 2026

Goal: Get interview-ready and start applying

Day 99 - March 24 (Monday)

Topic: Resume Creation

Tasks:

- ATS-friendly format
- Highlight projects with metrics
- Skills section (technical stack)
- Education section
- No fluff, concrete achievements

Example Bullet:

- "Built full-stack task manager (MERN) handling 1000+ tasks with JWT auth, deployed on Railway/Vercel"
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Day 100 - March 25 (Tuesday)

Topic: LinkedIn Optimization

Tasks:

- Professional headline
- About section (story-driven)
- Experience section (projects as experience)
- Skills endorsements

- Post about journey
 - Connect with Devsinc/Systems employees
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Day 101 - March 26 (Wednesday)

Topic: Interview Preparation - Technical

Practice:

- Common React interview questions
- JavaScript coding challenges
- Explain your projects
- System design basics
- Behavioral questions (STAR method)

Mock Interview:

- Record yourself answering questions
 - Practice whiteboard coding
 - Time yourself
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Day 102 - March 27 (Thursday)

Topic: Company Research & Applications

Target Companies:

1. Devsinc
2. Systems Limited
3. Arbisoft
4. Nisum
5. 10Pearls

Tasks:

- Research company culture
 - Find employees on LinkedIn
 - Tailor resume for each
 - Write cover letters
 - Apply through:
 - Company websites
 - LinkedIn
 - Referrals (best)
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Day 103 - March 28 (Friday)

Topic: Networking & Referrals

Activities:

- Message Devsinc employees (politely)
- Join Lahore tech communities
- Attend meetups (if available)
- Share projects in groups
- Ask for feedback

Script:

"Hi [Name], I'm a self-taught MERN developer from Lahore. I built [Hero Project] solving [problem]. Would love to learn about your experience at Devsinc. Here's my portfolio: [link]"

Day 104-105 - March 29-30 (Weekend)

Topic: Final Preparation & First Wave Applications

Tasks:

- Send 10 job applications
- Follow up on previous applications
- Practice coding challenges daily (LeetCode Easy)
- Prepare project demos
- Set up interview calendar

Success Metrics:

- Resume: ATS-optimized ✓
 - LinkedIn: Professional ✓
 - Portfolio: 5 deployed projects ✓
 - GitHub: Green squares ✓
 - Applications: 10 sent ✓
 - Network: 20+ connections in target companies ✓
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COMPLETE SKILLS MATRIX (BY WEEK 15)

FRONTEND SKILLS

- ✓ HTML5 (Semantic, Accessibility)
- ✓ CSS3 (Flexbox, Grid, Animations)
- ✓ Tailwind CSS (Utility-first styling)
- ✓ JavaScript ES6+ (Mastery level)

- ✓ React.js (Hooks, Context, Router)
- ✓ State Management (useState, useReducer, Context API)
- ✓ Form Handling (React Hook Form)
- ✓ API Integration (Fetch, Axios)
- ✓ Performance Optimization (React.memo, useMemo)

BACKEND SKILLS

- ✓ Node.js (Runtime, Modules)
- ✓ Express.js (REST APIs, Middleware)
- ✓ MongoDB (NoSQL database)
- ✓ Mongoose (ODM, Schemas, Validation)
- ✓ Authentication (JWT, bcrypt)
- ✓ File Upload (Multer)
- ✓ Security (CORS, Helmet, Env Variables)

TOOLS & WORKFLOW

- ✓ Git & GitHub (Version control, Collaboration)
- ✓ Chrome DevTools (Debugging, Performance)
- ✓ Postman (API testing)
- ✓ VS Code (Productive setup)
- ✓ npm/yarn (Package management)
- ✓ Deployment (Vercel, Railway, Netlify)

SOFT SKILLS

- ✓ Problem-solving (Systematic debugging)
 - ✓ Code documentation (README, Comments)
 - ✓ Time management (5-6 hour daily discipline)
 - ✓ Self-learning (AI tools, documentation)
 - ✓ Teaching-back (Explaining complex concepts)
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FINAL DELIVERABLES (PORTFOLIO)

8 DEPLOYED PROJECTS

1. **Weather Dashboard** (Week 3)

- Tech: Fetch API, localStorage, Geolocation
- Link: [Deployed URL]

2. **Movie Search App** (Week 4)

- Tech: OMDb API, Modal, Filters
- Link: [Deployed URL]

3. **Shopping Cart** (Week 5-7)

- Tech: React, Context API, Tailwind
- Link: [Deployed URL]

4. **E-Commerce Platform** (Week 9)

- Tech: React Router, Context + Reducer, Auth
- Link: [Deployed URL]

5. **Blog API** (Week 10)

- Tech: Express, Mongoose, JWT
- Link: [API Documentation]

6. **Task Manager MERN** (Week 12)

- Tech: Full MERN stack, Deployment
- Link: [Frontend + Backend URLs]

7. **Hero Project** (Week 13)

- Tech: Custom MERN application
- Link: [Deployed URL]

8. **Portfolio Website** (Week 14)

- Tech: React, Tailwind, Showcase
- Link: [Personal Domain]

GITHUB STATS

- Repositories: 15+
- Commits: 100+
- Stars: Target 10+ (from Hero Project)
- Green Squares: Consistent contributions

LINKEDIN PRESENCE

- Profile: Professional + Portfolio link
 - Posts: 5+ project showcases
 - Connections: 50+ (target companies)
 - Recommendations: Request from peers
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CRITICAL DATES & MILESTONES

Date	Milestone	Status
Dec 2, 2025	Training Start	✅ Complete
Dec 26, 2025	Week 3 Gatekeeper	✅ Passed (8.5/10)
Jan 4, 2026	Week 4 Start	🔄 Current
Jan 10, 2026	Phase 1 Complete	⌚ Upcoming
Jan 26, 2026	Week 6 Gatekeeper	⌚ Critical
Feb 16, 2026	Phase 2 Complete	⌚ Upcoming
Mar 9, 2026	Week 12 Gatekeeper	⌚ Critical
Mar 16, 2026	Hero Project Done	⌚ Upcoming
Mar 30, 2026	Training Complete	⌚ Target
Apr 1, 2026	Job Applications Start	⌚ Goal

EMERGENCY PROTOCOLS

IF UNIVERSITY EXAMS CONFLICT:

- Training pauses completely (like Dec 15-19)
- Resume exactly where left off
- No guilt, no pressure
- Exam dates adjusted in timeline

IF GATEKEEPER FAILED:

- Repeat entire week (no shortcuts)
- Identify specific gaps
- Additional practice problems
- Reattempt exam weekend
- Can fail max 2 times, then review entire phase

IF BURNOUT OCCURS:

- Mandatory 2-day break
- No coding during break
- Return with adjusted pacing
- Quality > Speed always

IF CONCEPT UNCLEAR:

- Stop immediately
 - Ask clarifying questions
 - Get 3 different explanations
 - Practice until clear
 - Never move forward confused
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